

1.

$\Pi_{\text{name}} (\sigma_{\text{topic} = 'AI' \wedge \text{type} = 'certificate' \wedge \text{duration} \leq 6} (\text{Course}))$

2. $\Pi_{\text{course.name}} (\sigma_{\text{topic} = 'AI' \wedge \text{type} = 'certificate' \wedge \text{duration} \leq 6} (\text{Course} \bowtie \text{Partnership} \bowtie \sigma_{\text{name} = 'IITKGP'(\text{University})}))$

3. $\Pi_{\text{student.name}} (\sigma_{\text{age} < 18 \vee \text{age} > 60} (\text{Student} \bowtie \text{Enrollment} \bowtie \sigma_{\text{name} = 'GenAI'(\text{Course})}))$

4. $\Pi_{\text{student.name}} (\sigma_{\text{country} \neq 'India'} (\text{Student} \bowtie \text{Enrollment} \bowtie \text{Course} \bowtie \text{Partnership} \bowtie \sigma_{\text{name} = 'IITKGP'(\text{University})}))$

5. $\Pi_{\text{country}} (\text{Student} \bowtie \text{Enrollment} \bowtie \text{Teaches} \bowtie \sigma_{\text{name} = 'Andrew'(\text{Instructor})})$

6. $\Pi_{\text{instructor.name}} (\text{Instructor} \bowtie \text{Teaches} \bowtie \text{Enrollment} \bowtie \sigma_{\text{country} = 'India'} (\text{Student}))$

7. $\Pi_{\text{c.name}} (\text{Enrollment } E1 \bowtie \text{Enrollment } E2 \wedge (E1.\text{student-id} = E2.\text{student-id} \wedge E2.\text{course-id} = \sigma_{\text{name} = 'GenAI'(\text{Course})}))$

8. $\Pi_{\text{course-id}} (\text{Enrollment}) - \Pi_{\text{course-id}} (\text{Enrollment} -$

$\Pi_{\text{student-id, course-id}}$

$(\text{Enrollment} \bowtie \sigma_{\text{name} = 'GenAI'(\text{Course})})$

9. $\gamma_{\text{course-id}; \text{count}(\text{student-id}) \rightarrow \text{cnt}} (\text{Enrollment} \bowtie \text{Partnership} \bowtie \sigma_{\text{name} = 'IITKGP'(\text{University})})$
(select max cnt)

10. $\gamma_{\text{course-id}; \text{avg}(\text{marks}) \rightarrow \text{avgm}} (\sigma_{\text{country} = 'India'} (\text{Student} \bowtie \text{Enrollment} \bowtie \sigma_{\text{topic} = 'AI'(\text{Course})}))$
(select max avgm)